

Supplementary Information — Junction gapmers across the NR4A3 fusions of extraskelatal myxoid chondrosarcoma

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Research use only, and not for administration to any person or animal. Every oligonucleotide sequence named in this Supplementary Information, in the main text and in their tables is a research reagent intended solely for laboratory investigation. None is a medicine or a candidate drug, none has been synthesised or tested by anyone, and none may be administered to any human being or animal, compounded for such use, or supplied to any person for such use. Custom oligonucleotide synthesis is commercially available, so the restriction is on use rather than on access. The full statement is in the main text's Declarations.

Section numbers here are prefixed S. Cross-references of the form "`§n`" point to the main text (`fusion-junction-aso-research-article.md`); "`SI §Sn`" points within this document. All numbered references are those of the main text's reference list, which is generated from the identifiers carried in the main text. This document carries no figure: Supplementary Figure S1 travels with the archive, and its legend is with the main text's figure legends.

S1 · Target-site accessibility, and its exclusion from every ranking

§6 gives the estimator, the range across the panel and the fact that nothing in the main text is ranked on it. Here are the three reasons for that omission, in decreasing order of force.

First, the quantity is not the one the paper is about: accessibility predicts whether an oligonucleotide can reach its site on the fusion, and every question asked in the main text is whether it can be told apart from a parent once it does. An inaccessible site is a potency problem, and potency is not claimed for any sequence.

Second, the estimate is a fold of a naked transcript, whereas the compartment that matters is a nascent, protein-coated pre-mRNA; measured antisense activity correlates with such predictions weakly and inconsistently, which is why the field selects by screening rather than by folding.

Third, it would have no purchase on the result even if it were reliable: the surviving candidates are separated by orders of magnitude of predicted off-target load, and reordering them on a quantity that spans 0.160 to 0.707 across the panel would substitute a weak predictor for a strong one. The values are released so that a laboratory ordering several oligonucleotides at one junction can break a tie on them, which is the use they support.

S2 · The melting-temperature cross-check on the duplex thermodynamics

The 250 nM strand concentration used in the main text's thermodynamic scoring enters only the melting temperature, which is the quantity the independent implementation of §6 was compared against. The strand choice that check does not verify is fixed instead by the documented convention of the nearest-neighbour table⁵² that the sequence supplied is the RNA one.

S3 · Provenance of the three gap-length figures

§6 states the three figures that place a six-nucleotide gap below the reported optimum, and records that none of them is a titration in this architecture. Their provenance is as follows.

- **A minimum of five.**⁴¹ Given in passing, citing prior work rather than measuring it.
- **A minimum of six.**⁴² A design-protocol statement, likewise given in passing and citing prior work.
- **An optimum of seven to ten.**³⁸ Both this and the six-nucleotide activity figure beside it are taken from a review that credits them to named earlier primary studies rather than measuring them itself.

S4 • The graded re-score: which screens carry it

The re-score of §6 covers all 38 junction screens, and 39 of the 93 screens released in total. Two classes of screen are released ungraded. One coverage-only control screen records no gap-mismatch depth and so cannot be graded at all; the 53 deeper re-screens are released ungraded because the graded model adds nothing where no hit list is truncated. The re-score counts only sense-strand hits where it can, meaning where the retained hit list is complete and every hit's strand is therefore known.

Which of the two bounds applies to a given design follows from its hit list, and each design records that. Screens produced before the orientation fix carry a strand-blind count for every truncated design, because the strand of an unretained hit is unrecoverable; screens produced afterwards carry an already-filtered one.

S5 • The two unfiltered control screens

The two screens §6 records as unfiltered are modelled control junctions built in amino-acid rather than transcript coordinates. They carry no junction from the 38-junction panel, and no count reported anywhere in the main text is taken from them.

S6 • The coverage ladder's second basis, and four zero-contribution junctions

Table 5 prices every rung on the breakpoint distribution of a single 18-case series,²² which is the basis 68.4% is computed on. Every figure in this section is arithmetic over published cohorts rather than a coverage measurement — no patient was screened with any sequence named here — and §4.1 states in full what that figure is, what it is not, and the four bounds on it. Two things sit around that ladder: where its arithmetic runs out, and a second basis that answers a different question and supersedes nothing.

The arithmetic runs out before the target does. A set covering every *EWSR1* and every *TAF15* breakpoint reaches 94.8% of molecularly confirmed cases and stops, so no panel restricted to those two partners reaches 95%, and the remaining reachable cases are the two *TCF12* tumours that the fourth reagent of §4.1 addresses.

Nine junctions now carry a published exon-resolved breakpoint and a screened design, eight of them through all five screens, with the *PGR* seam's pre-mRNA compartment unmeasured for the reason main text §2.6 gives, and four of them sit outside the 38-junction panel, at the *NR4A3* exon-2 acceptors of §2.6. Four of the nine move the estimate; one moves only the bound, its arm having no measured within-partner distribution; and four move it by nothing at all — two because their partner is absent from this cohort, and two because their partner is present while their exon pair carries no count in it. That last pair is the easier of the two zeroes to miss, and a membership test that asked only about the partner did miss it. Priced on a pooled breakpoint basis rather than on the single series the ladder uses, the nine together are 82.9% of molecularly confirmed cases, widening to 57.5–90.7%, and the two reagents of §4.1 are 67.1% rather than the ladder's 68.4%. The denominators are not the ladder's, and neither are the numerators: the *EWSR1* arm is 17 of 20 here against the ladder's 10 of 15, and two changes make that move rather than one. The denominator widens by pooling the 18-case series with the five-case whole-transcriptome cohort,²⁵ and the numerator rises because this panel carries four *EWSR1* junctions where the ladder carries one, reaching 12 of the

same 15 tumours before any pooling. The *TAF15* arm stays at 3 of 3. Three things follow. Fifteen of those 20 tumours come from one series, so the pool stays close to it; the two series agree at *EWSR1* exons 12 and 13 and disagree completely at exon 7 to *NR4A3* exon 2, 0 of 15 against 1 of 5; and a third series was refused because every case in its *EWSR1* arm carries a covered junction by construction, which would fix its rate at 100% and raise the figure rather than lower it. Two of seven retrieved series resolve every case to an exon pair, so this is a pooled record and not the whole one. The figure supersedes nothing; it answers a different question from the ladder, whose rungs are incremental and priced on the single series 68.4% is computed on, and 68.4% remains the coverage of the two reagents named in §4.1. Its own membership rule is an evidence test rather than a list — a junction qualifies where a published report places a patient's breakpoint at it and a reagent has been through all five screens, four of the five for the *PGR* seam whose pre-mRNA compartment is unmeasured rather than clean (main text §2.6) — and one qualifying junction, *PGR* exon 2 to *NR4A3* exon 2, reported in a single patient,²⁸ moves the figure by exactly zero, because the 58-case cohort behind the denominator contains no *PGR* case for such a reagent to engage. The further caveat on that reagent — a sixth partner, outside the five the panel models, whose seam is screened against the non-canonical-acceptor table rather than through the panel's transcript models — is in §2.6, which also points at the accession §6 lists for it. *TFG* exon 7 to *NR4A3* exon 3 is the second of that pair, on the same ground and from a deposited cDNA rather than a report. Those two are the partner-absent half of the four that add nothing; the other two have their partner present in the cohort and their exon pair carrying no count in it. What such a reagent changes is which patients are reachable at all, which is a different statement and is not added to a coverage percentage.